Wing Flaps System – General Data

1. General

Refer to Figure 1 and Figure 2 for details.

The wing flaps system consists of inboard and outboard flaps hinged to each wing; each outboard flap has an aileron hinged at its trailing edge and each inboard flap has a trailing flap hinged at its trailing edge. A selector lever in the flight compartment controls a hydraulic selector valve, which determines the position of a hydraulic actuator. The actuator, through linkage, bellcranks and push-pull rods, connects to and operates the wing flaps. An indicator system is provided to register the position of the flaps. Lights are provided on each side of the flap selector lever and inside the flap position indicator pointer (refer to 33-10-00, Flight Compartment Lights (Mod 6/2028) – General Data.

2. Description

The inboard (fore and trailing) and outboard flaps hinged to each wing are all interconnected by push-pull rods, idler levers and bellcranks, which are connected to a single hydraulic actuator piston rod by links. The flap selector lever in the flight compartment, is connected by cables to a selector pulley assembly mounted on the selector valve operating shaft; the selector valve being attached to the flap actuator. The selector pulley assembly comprises two pulleys mounted side by side, either of which operates the selector valve independently. The cables from the flap selector lever are connected to the outboard pulley, and a follow-up cable system from the flap actuator piston rod connects to the inboard pulley. Hydraulic pipe lines from the aircraft main hydraulic system (refer to 29-00-00, Hydraulic Power – General Data) connect to the flap selector valve. Flap position indication is provided on the Multi-Function Display. To maintain aircraft trim during flap operation, an interconnecting system to the right elevator trim tab is installed.

A. Wing Flaps

The inboard and outboard flap hinge arms, are hinged to the flap hinge arm attachment brackets, which are bolted to the wing structure. A common flap hinge arm attachment bracket is located at station 172.50 and accommodates the end hinge arms of the inboard fore and outboard flaps. Fore flap hinge arms at stations 97.50 and 247.15 (approximately) are connected to the flap system push-pull rods. The inboard trailing flap hinge arms are hinged to those of the fore flap, and a connecting rod connects both flap hinge arms at station 97.50 to control the travel of the trailing flap.

B. Wing Flap Hydraulic System

The wing flap hydraulic circuit lines are tapped into the main hydraulic system lines at the hydraulic power package located beneath the flight compartment floor, and are routed to the flap selector valve on the flap actuator. Restrictions are provided in the flaps up and flaps down ports of the actuator and a check valve is fitted in the pressure supply line at the flap selector valve. Both the flap up and down lines, incorporate a thermal relief valve.

C. Flap Actuator

The wing flap actuator, with the selector valve attached, is mounted in a frame located centrally in the cabin roof. The actuator piston rod is connected by links to the left and right operating bellcranks.

D. Flap Selector Lever

The flap selector lever assembly, mounted in the flight compartment overhead console, comprises a selector lever, quadrant, pulley and cable system, and serrated segment. The selector lever, located behind the engine control levers, is connected by a connecting rod to the quadrant immediately forward of the lever. A pawl on the lever, engages the serrated segment about which the lever pivots, for intermediate settings of the flaps. Cables connected to the quadrant are routed over pulleys to connect to the outboard pulley on the flap selector valve spindle.

E. Flap Follow-up System

A cable connected to a bracket on the actuator piston rod, is routed to the inboard pulley on the flap selector valve spindle.

F. Flap Position Indicator

The flap position indicator system comprises of a rotary potentiometer installed on the LH flap. As the flap is extended, the transducer provides an analog electrical output signal (0 to 28 Volt DC) to the Data Acquisition Units. Using an interface (ARINC429) the signal is converted and flap position is shown on the EFIS computer screen in the flight compartment. Flap position indication is provided on the Multi-Function Display in the flight compartment.

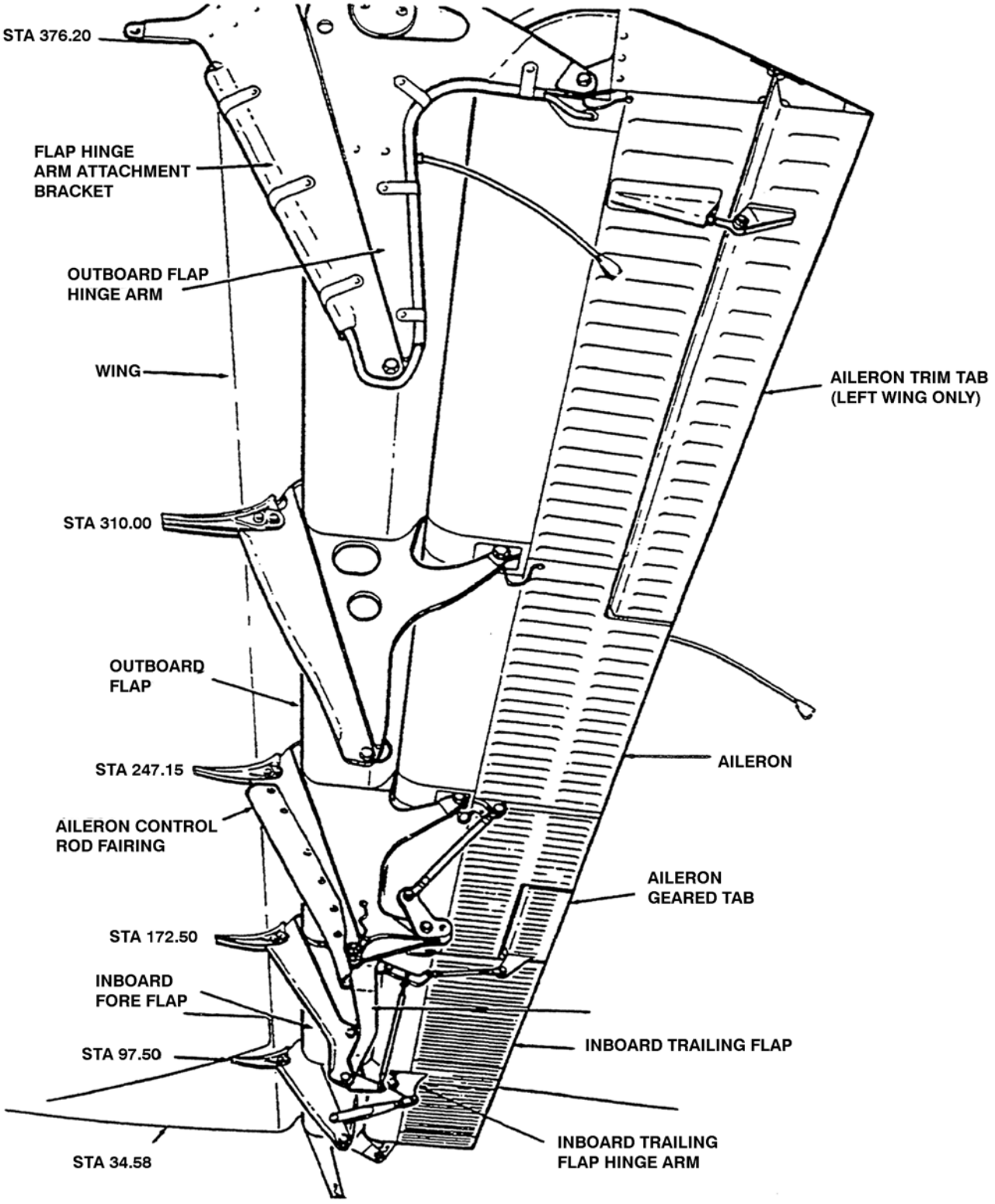
3. Operation

The wing flap hydraulic system derives its power for the wing flap actuator from the aircraft hydraulic system main pressure line, and returns displaced fluids from the actuator into the aircraft hydraulic system main return line (refer to 29-00-00, Hydraulic Power – General Data). When the aircraft hydraulic system is pressurized, and the flap selector lever is moved to the flaps down position, a corresponding amount of movement is transmitted to the quadrant, and through the cables to the outboard pulley on the selector valve shaft. Rotation of the outboard pulley opens the selector valve to permit hydraulic fluid under pressure to enter the flap actuator down port to extend the actuator piston rod. The piston rod, through the links, operates the left and right inboard operating bellcranks, which through the push-pull rods, idler levers and outboard operating bellcranks, moves the flaps to the down position. During actuator piston rod extension, the follow-up mechanism cable rotates the inboard cable pulley on the selector valve shaft, which is part of an epicyclic gear train, to close the selector valve. The reverse of the above operation applies when a flaps up selection is made.

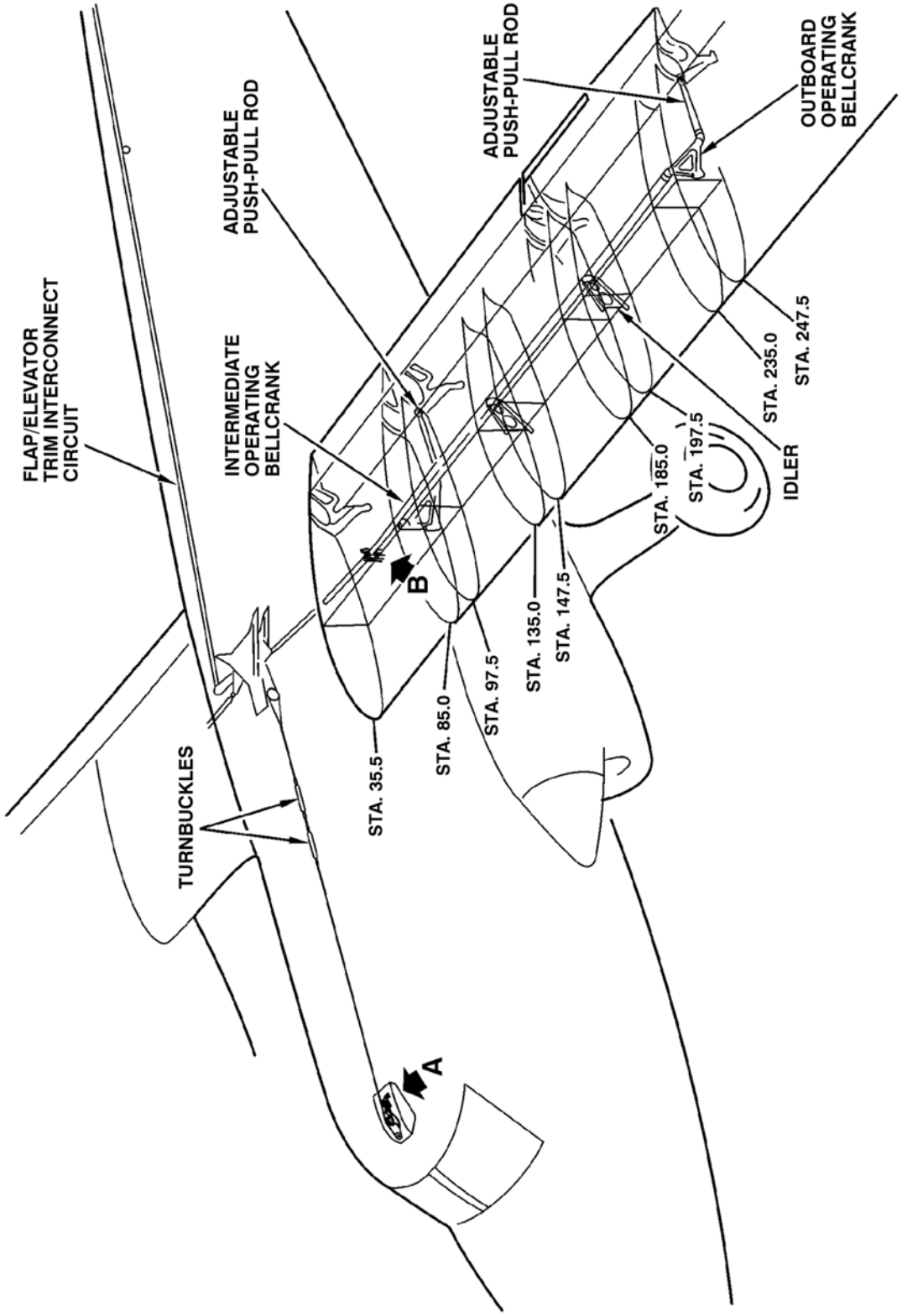
The Multi-Function Display (MFD) registers the flap position by the signal it receives from the position transducer installed on the LH flap.

A flap/elevator trim interconnect ball screw jack, attached to the airframe structure and the right inboard operating bellcrank, operates during flap movement to transmit movement to the flap/elevator trim interconnect circuit to maintain aircraft elevator trim during flap operation.

The restrictors in the actuator flaps up and down ports limit the rate of flap movement, and the check valve in the pressure supply line isolates the flap circuit in the event of a failure in the hydraulic pressure supply.

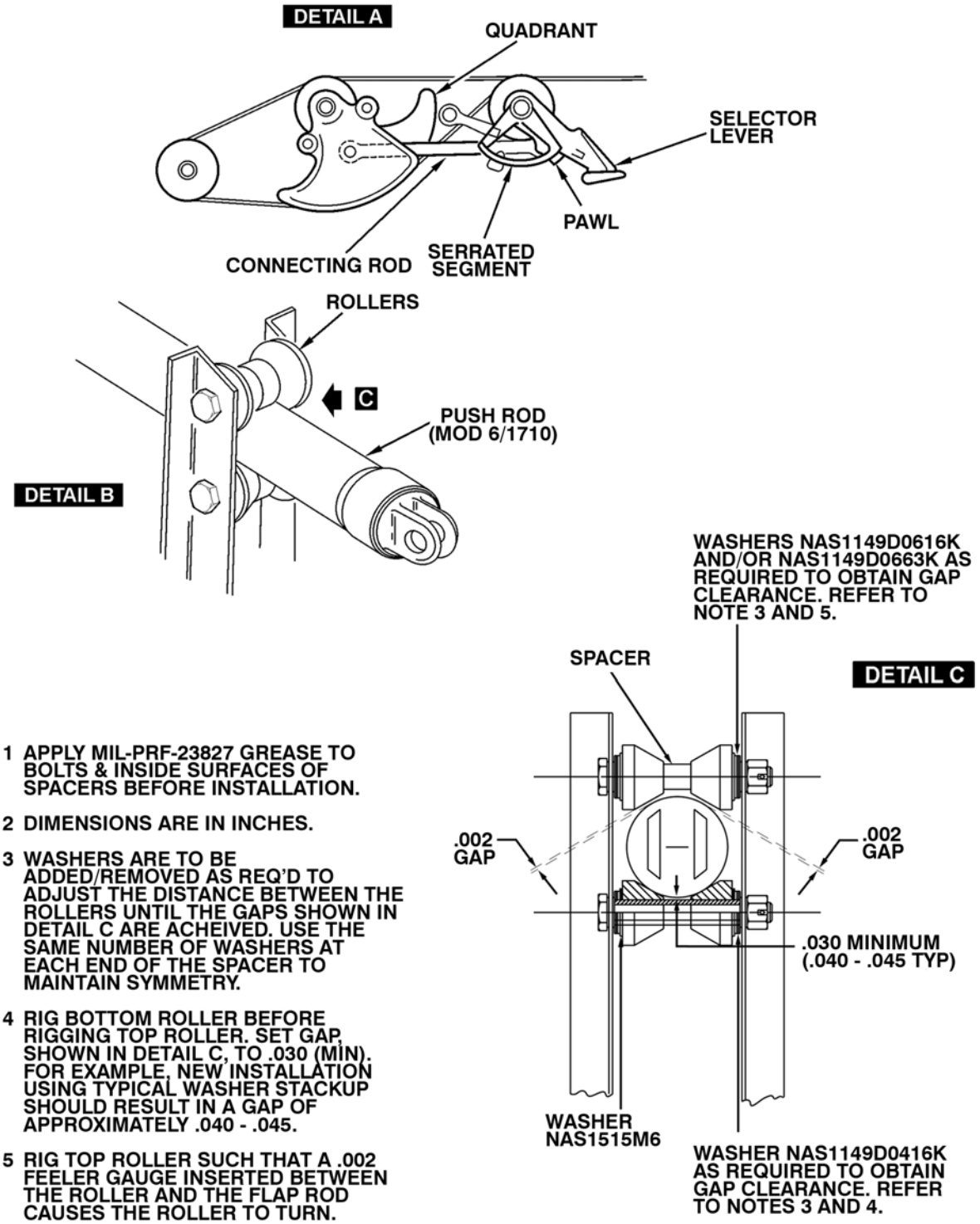


Wing Flap Arrangement
Figure 1



Wing Flap System
Figure 2 (Sheet 1 of 4)

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NOTE: 1 APPLY MIL-PRF-23827 GREASE TO BOLTS & INSIDE SURFACES OF SPACERS BEFORE INSTALLATION.

2 DIMENSIONS ARE IN INCHES.

3 WASHERS ARE TO BE ADDED/REMOVED AS REQ'D TO ADJUST THE DISTANCE BETWEEN THE ROLLERS UNTIL THE GAPS SHOWN IN DETAIL C ARE ACHEIVED. USE THE SAME NUMBER OF WASHERS AT EACH END OF THE SPACER TO MAINTAIN SYMMETRY.

4 RIG BOTTOM ROLLER BEFORE RIGGING TOP ROLLER. SET GAP, SHOWN IN DETAIL C, TO .030 (MIN). FOR EXAMPLE, NEW INSTALLATION USING TYPICAL WASHER STACKUP SHOULD RESULT IN A GAP OF APPROXIMATELY .040 - .045.

5 RIG TOP ROLLER SUCH THAT A .002 FEELER GAUGE INSERTED BETWEEN THE ROLLER AND THE FLAP ROD CAUSES THE ROLLER TO TURN.

6 BOLTS SHOULD NOT BE OVER TIGHTENED. ROLLERS/SPACERS SHOULD TURN FREELY ON BOLTS.

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Wing Flap System
Figure 2 (Sheet 2 of 4)

NOTE: 1 APPLY MIL-PRF-23827 GREASE TO BOLTS & INSIDE SURFACES OF SPACERS BEFORE INSTALLATION.

2 DIMENSIONS ARE IN INCHES.

3 WITH ROD PUSHED ALL THE WAY INSIDE THE WING (I.E. FLAP FULLY RETRACTED). PRE-FIT ROLLERS, SPACERS, & BOLTS (NOTE: NO WASHERS) BY CENTERING THE ROD ON ROOT RIB CUT-OUT AND BETWEEN ROLLERS/SPACERS. SPACERS SHALL BE APPROX CENTERED BETWEEN SUPPORT ANGLES.

4 SLIDE ROLLERS TO MAKE CONTACT WITH ROD, THEN MEASURE GAPS G1, G2, G3 & G4. REMOVE PRE-FIT ROLLERS, SPACERS & BOLTS.

5 DETERMINE QUANTITY OF NAS1149D0663K AND NAS1149D0616K WASHERS USING DIMENSION: Gx-.05 = Tx; CHOOSE EQUAL OR NEXT GREATER THICKNESS.

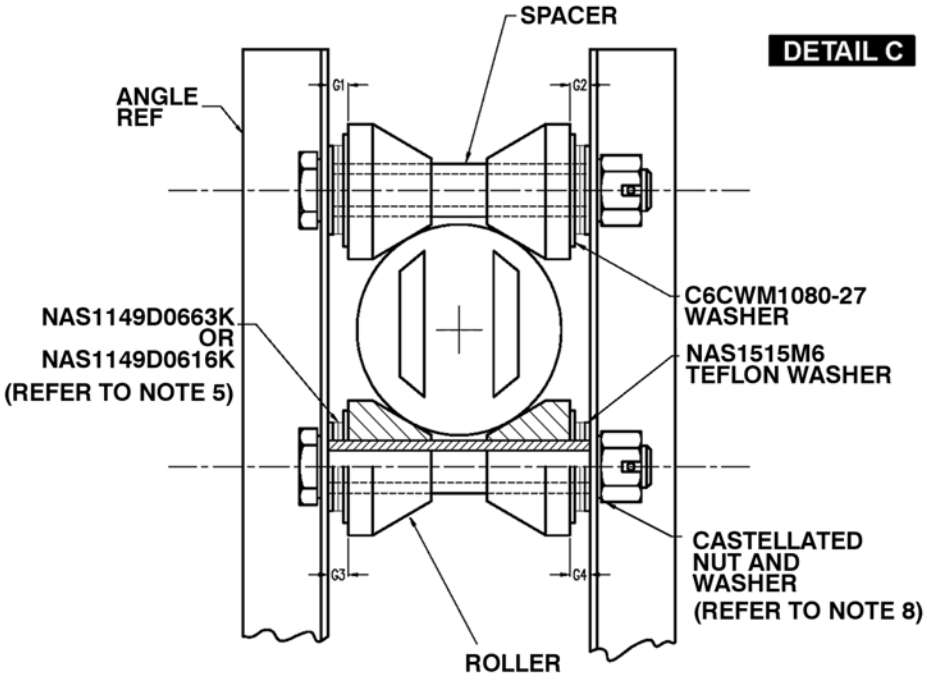
EXAMPLE:
G1=.189, THEREFORE G1-.05 = .139
REQD WASHER: 2x NAS1149D0663K = .126
1x NAS1149D0616K = .016
.142

6 INSTALL BOTTOM ROLLER/SPACER WITH WASHERS AS PER NOTE 5. ENSURE THE SPACER IS APPROX CENTERED: I.E. EQUAL GAP BETWEEN FWD & AFT SUPPORT ANGLE.

7 WHILE PUSHING THE FLAP ROD DOWNWARD TO COMPRESS THE LOWER ROLLERS, INSTALL THE UPPER ROLLERS/SPACER AS PER NOTE 5. CENTER THE SPACER ON THE FLAP ROD.

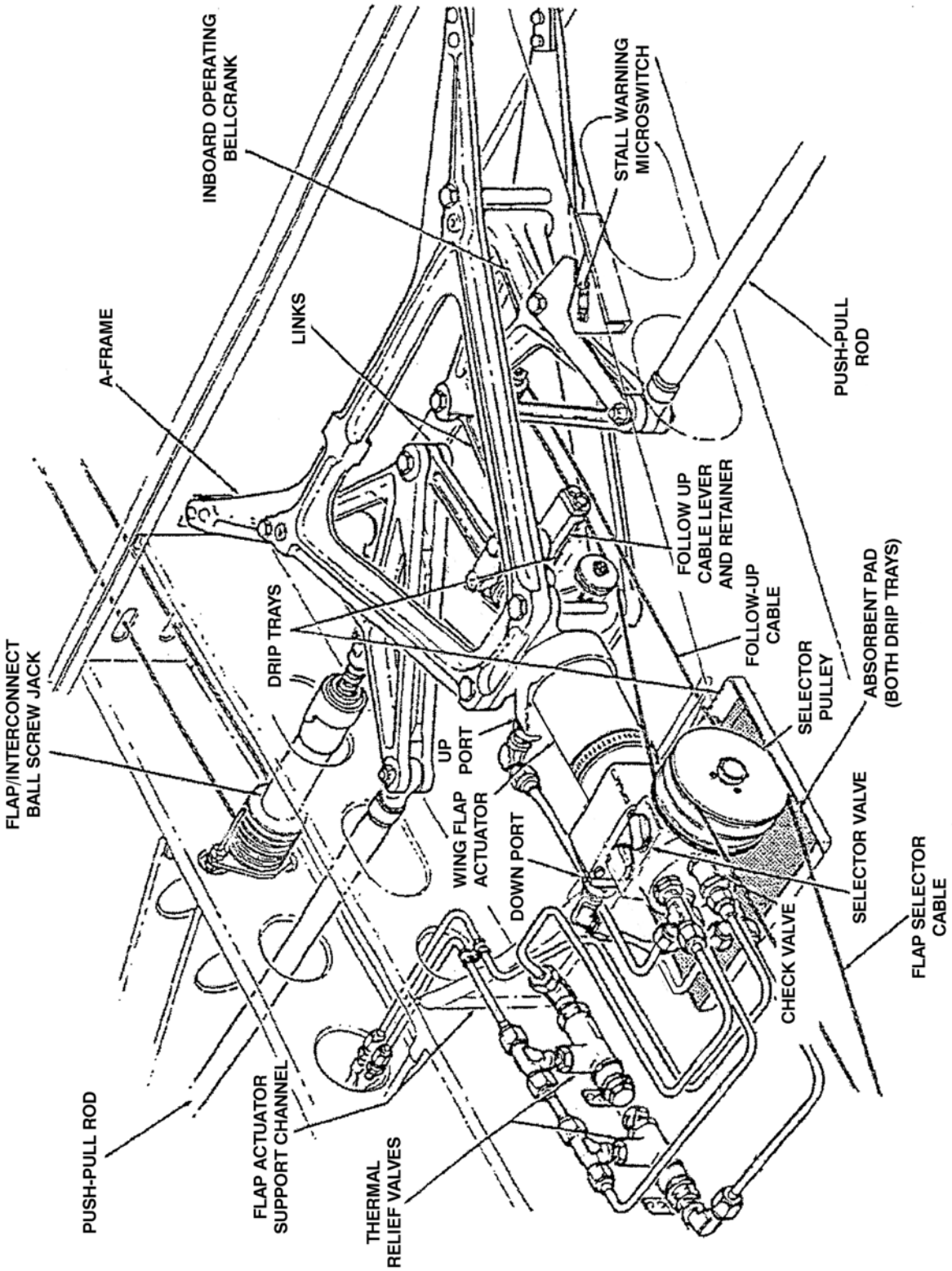
8 FIRST, TIGHTEN BOTH NUTS TO ENSURE A SNUG CONTACT AGAINST THE WASHER/BRACKET. THEN, TIGHTEN BOTH NUTS BY 1 TO 3 CASTELLATED SLOTS TO ALIGN COTTER PIN HOLE ON THE BOLT. IF MORE THAN 50% OF PIN HOLE IS ABOVE THE NUT CASTELLATION, ADD NAS1149D0463K OR NAS1149D0416 WASHER UNDER NUT.

9 MAKE SURE THE ROLLERS/SPACERS ARE TURNING FREELY ON BOLTS (ROD NOT SLIDING) WHILE THE FLAP RODS MOVE IN AND OUT.



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Wing Flap System
Figure 2 (Sheet 3 of 4)



Wing Flap System
Figure 2 (Sheet 4 of 4)

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4. Servicing

A. Bleed Wing Flap System

Note Bleeding the wing flap system need only be accomplished if the flap operation is not smooth, a hydraulic line or component has been replaced, or the main hydraulic system has been dismantled.

1) Check that hydraulic reservoir is full.

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), and check that BUS TIE switch is set to NORMAL.

3) Set HYDRAULIC OIL PUMP circuit breaker.

Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1575 ± 50 psi (Mod 6/1570).

4) Check that flap area is clear of equipment and personnel.

5) Select flaps down and allow them to travel to full down position, select flaps up and allow them to travel to full up position. Repeat for ten full selections to ensure complete bleeding of system.

6) Check flap system for smooth operation.

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

5. Adjustment/Test

Refer to Figure 3 for details.

A. Rig Wing Flaps

1) Move flap selector lever to flaps up position, using handpump; pump flaps to fully up position.

2) Check that cable terminals on flap selector and follow-up pulley are 30° from the vertical and that the distance between each terminal, measured in a horizontal plane fore and aft, is 1.75 inches.

3) If cable terminal settings are incorrect, remove attachment bolts and disconnect the left and right push-pull rods at each inboard operating bellcrank. Support flaps in down position and carry out Para 5.A.4) through Para 5.A.26) as necessary.

4) With flap selector in up position (against rear stop), operate handpump to fully retract flap actuator.

5) Slacken two 3/8–inch bolts attaching bellcrank links to flap actuator piston rod sufficiently to loosen follow-up cable lever.

6) Slacken selector cable turnbuckles in cabin roof, and set cable terminal in outboard pulley to 30° forward of vertical; check dimension between terminals, measured fore and aft, is 1.75 inches.

7) Adjust selector cable turnbuckles in cabin roof to tension cable to values given in 27-00-00, Flight Controls – General Data. Ensure that settings made to both pulleys remain unchanged.

8) Tighten two 3/8–inch bolts loosened in Para 5.A.5), and adjust follow-up cable adjusting nut on lower "A" frame to tension cable to value given in 27-00-00, Flight Controls – General Data, ensuring that settings on both pulleys remain unchanged.

9) Move selector to down position (against serrated stop) and operate handpump until actuator piston has extended exactly 3.40 inches. Adjust serrated stop to limit movement of selector lever exactly at this point.

Note If movement of 3.40 inches cannot be obtained, the follow-up valve is misaligned and Para 5.A.5) through Para 5.A.8) must be repeated to ensure correct follow-up valve alignment before proceeding.

10) Retract and extend actuator several times to ascertain piston extension is constant at 3.40 inches.

11) Connect push-pull rods disconnected in Para 5.A.3).

12) Move selector lever to up and pump flaps fully up.

13) Check that trailing edge of inboard trailing flaps align with rigging plates on each side of fuselage and rigging plates on inboard and outboard fore flaps are aligned with wing shroud trailing edge.

14) If fore flaps are not aligned with their rigging plates, remove attachment bolt and disconnect adjustable push-pull rod at each inboard and outboard fore flap on both sides and carry out Para 5.A.15) through Para 5.A.26).

15) Screw out, to full extension, push-pull rod to left inboard fore flap and reattach to flap.

16) Pump flap to full up and adjust push-pull rod, as necessary, until rigging plate is aligned.

17) Repeat Para 5.A.15) and Para 5.A.16) for right inboard fore flap, then left and right outboard fore flaps, in that order.

18) Check that all fore flap hinge arms are bearing equally on adjacent rubber pads when flaps are fully up, and a gap of 0.040/0.200 inch exists between fore flap hinge arms and flap hinge arm attachment brackets.

Note If required, trim rubber pads to suit actual gap after rigging.

19) If, with fore flaps fully up, inboard trailing flaps are not aligned with fuselage rigging plates, remove attachment bolt, and disconnect connecting rod of each trailing flap and carry out Para 5.A.20) and Para 5.A.21).

20) Hold left inboard trailing flap up in line with fuselage plate and adjust connecting rod until attachment bolt can be freely installed.

21) Repeat Para 5.A.20) for right inboard trailing flap.

22) Check left and right aileron rigging (refer to 27-10-00, Aileron Control System – General Data).

23) Ensure that all push-pull and connecting rod ends are tightened and correctly locked.

24) Check all flap control surface clearances are within limits (refer to 27-00-00, Flight Controls – General Data).

25) Operate flaps and check for full and free range of movement.

26) Check rigging of flap/elevator interconnect trim system (refer to 27-36-00, Flap/Elevator Interconnect Trim System – General Data).

27) Do a flap position potentiometer function test as per 27-50-31, Flap Angle Position Indicator – General Data, Para 3.B.

28) If adjustment is required for the flap position potentiometer, adjust as per 27-50-31, Flap Angle Position Indicator – General Data, Para 3.A.

B. Operational Test Wing Flap System

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), and check that BUS TIE switch is set to NORMAL.

2) Set HYDRAULIC OIL PUMP circuit breaker.

   Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1550 ± 50 psi.

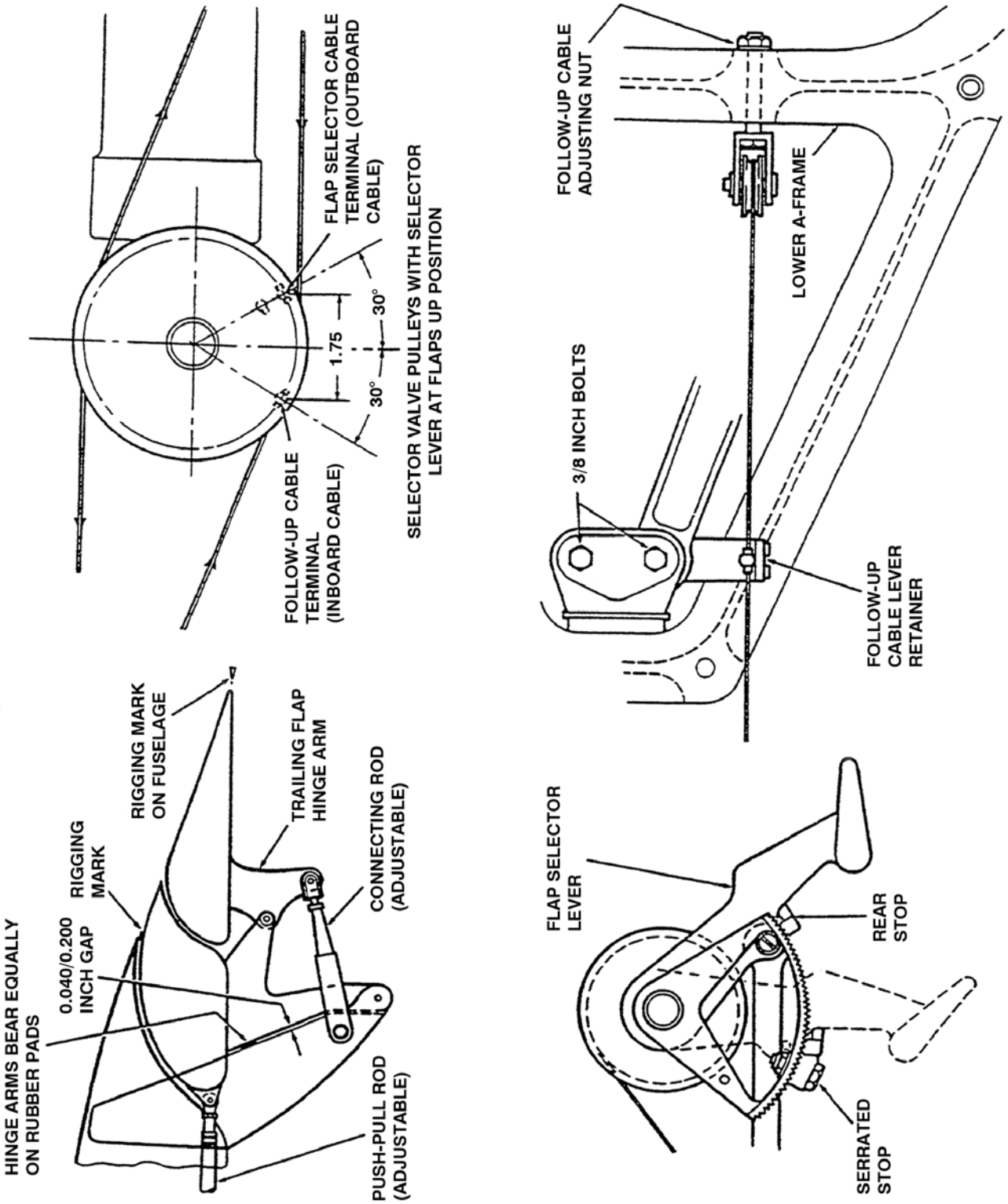
3) Check that flap area is clear of equipment and personnel.

4) Select flaps down and check that flaps are fully extended in 19 to 25 seconds.

5) Select flaps up and check that flaps are fully retracted to 35 to 50 seconds.

6) Make several selections and check for smooth operation over full travel.

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).



Wing Flaps System – Rigging
Figure 3

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